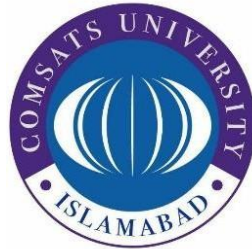


COMSATS University Islamabad, Attock Campus

Department of Computer Science



Course:

ARTIFICIAL INTELLIGENCE

Lab Project

Smart Travel Expert System

Submitted To

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Project Report:

Smart Travel Expert System

1. Problem Description

The Smart Travel Management Expert System is a rule-based expert system developed in SWI-Prolog to assist users in planning trips intelligently. Many travelers face difficulties in selecting suitable destinations, checking visa requirements, estimating travel costs, finding safe places, and choosing accommodations according to their budget and preferences.

This system solves these problems by using Artificial Intelligence techniques, specifically a knowledge-based expert system, where facts and rules are stored in a Prolog knowledge base. The system analyzes user preferences such as travel type (business/leisure), budget (affordable/moderate/premium), destination safety, weather conditions, and other travel-related information to provide intelligent recommendations.

The project demonstrates how AI can automate decision-making in the travel and tourism domain through logical inference and rule-based reasoning.

2. Objectives

The main objectives of the Smart Travel Management Expert System are:

- To develop an AI-based expert system using SWI-Prolog.
- To provide intelligent travel recommendations according to user preferences.
- To help users estimate travel costs including transport and accommodation.
- To check visa requirements for different destinations.
- To suggest safe travel destinations.
- To provide information about hotels, famous foods, festivals, and weather risks.
- To demonstrate the use of rule-based inference in solving real-world problems.
- To create an interactive chatbot-style interface for users.

3. Modules of System

- User Management
Stores user information such as travel type, budget, and country.
- Destination Recommendation
Suggests suitable travel destinations according to budget and travel type
- Hotel Management
Manages hotel and accommodation recommendations for travelers.
- Visa Checking
Checks whether a visa is required for a selected destination.
- Budget Planning
Estimates total travel expenses including transport and accommodation costs.
- Safety Advisory
Provides safety information and travel risk alerts for destinations.
- Food Recommendation
Suggests famous foods and cuisines of different destinations.
- Travel Insurance
Provides information about travel insurance requirements.
- Itinerary Planning
Generates day-wise travel plans and activity schedules for users

4. Knowledge Base

- Destination facts (type, budget, weather, speciality)
- Hotel database (affordable/premium)
- Visa rules for Pakistani travelers
- Weather risk information
- Safety ratings
- Famous food data
- Festivals and events
- Special-purpose destinations (honeymoon, adventure, etc.)

5. Rules / Inference Engine

The Rules and Inference Engine are the intelligent part of the expert system. They apply logical rules on the knowledge base to generate decisions, recommendations, and outputs automatically. The inference engine matches user preferences with stored facts and produces suitable travel suggestions.

INFERENCES

```
45 ?- learn_preference(leisure, moderate), recommend_trip(X).
X = bali .

46 ?- adventure_trip(X).
X = thailand.

47 ?- check_visa(dubai, X).
X = 'Visa Required'.

48 ?- estimate_cost(skardu, X).
X = 10000.

49 ?- safe_destination(X).
X = dubai .

50 ?- famous_food(Paris, X).
X = french_pastries.

51 ?- best_rated(X).
X = dubai .

52 ?- convert_currency(100, usd, X).
X = 28000.

53 ?- itinerary(dubai, Day, Activity).
Day = day1,
Activity = shopping .

54 ?- weather_risk(X, Y).
X = thailand,
Y = 'High Monsoon Risk' .
```

6. Queries and Outputs

IMPLEMENTATION

```
43 ?- consult(['c:/Users/samri/OneDrive/Documents/AI_LAB/PROLOG/travel_management.pl']).
true.

44 ?- start.
=====
SMART TRAVEL MANAGEMENT SYSTEM
=====

----- MAIN MENU -----
1. Recommend Trip
2. Visa Check
3. Cost Estimate
4. Safe Destinations
5. Famous Food
6. Best Rated Destinations
7. Emergency Contacts
8. Exit
-----
Enter choice: 1.

=== TRIP RECOMMENDATION ===
Enter trip type (business/leisure): |: leisure.
Enter budget (affordable/moderate/premium): |: moderate.

>>> Recommended Destination: bali
```

```

----- MAIN MENU -----
1. Recommend Trip
2. Visa Check
3. Cost Estimate
4. Safe Destinations
5. Famous Food
6. Best Rated Destinations
7. Emergency Contacts
8. Exit
-----
Enter choice: |: 2.

=== VISA CHECK ===
Enter destination name: |: dubai.

>>> Visa Required

----- MAIN MENU -----
1. Recommend Trip
2. Visa Check
3. Cost Estimate
4. Safe Destinations
5. Famous Food
6. Best Rated Destinations
7. Emergency Contacts
8. Exit
-----
Enter choice: |: 3.

=== COST ESTIMATE ===
Enter destination name: |: skardu.

>>> Estimated Cost (PKR): 10000

----- MAIN MENU -----
1. Recommend Trip
2. Visa Check
3. Cost Estimate
4. Safe Destinations
5. Famous Food
6. Best Rated Destinations
7. Emergency Contacts
8. Exit
-----
Enter choice: |: 4.

=== SAFE DESTINATIONS ===
- dubai
- skardu
- london
- bali
- maldives
- switzerland
- singapore

----- MAIN MENU -----
1. Recommend Trip
2. Visa Check
3. Cost Estimate
4. Safe Destinations
5. Famous Food
6. Best Rated Destinations
7. Emergency Contacts
8. Exit
-----
Enter choice: |: 5.

=== FAMOUS FOOD ===
Enter destination name: |: paris.

>>> Famous Food: french_pastries

----- MAIN MENU -----
1. Recommend Trip
2. Visa Check
3. Cost Estimate
4. Safe Destinations
5. Famous Food
6. Best Rated Destinations
7. Emergency Contacts
8. Exit
-----
Enter choice: |: 6.

=== BEST RATED DESTINATIONS (4.5 and above) ===
- dubai : 4.5
- bali : 4.7
- switzerland : 4.9
- maldives : 4.8
- paris : 4.6
- singapore : 4.7
- london : 4.5

----- MAIN MENU -----
1. Recommend Trip
2. Visa Check
3. Cost Estimate
4. Safe Destinations
5. Famous Food
6. Best Rated Destinations
7. Emergency Contacts
8. Exit
-----
Enter choice: |: 7.

=== EMERGENCY CONTACTS ===
Enter destination name: |: paris.

>>> Emergency Contact: emergency_112

```

```
----- MAIN MENU -----  
1. Recommend Trip  
2. Visa Check  
3. Cost Estimate  
4. Safe Destinations  
5. Famous Food  
6. Best Rated Destinations  
7. Emergency Contacts  
8. Exit
```

```
-----  
Enter choice: |: 8.
```

```
=====
```

```
    Thank You for Using Smart Travel  
    Expert System!
```

```
=====
```

```
true .
```

7. Conclusion

The Smart Travel Management Expert System is a rule-based AI system developed using SWI-Prolog that helps users plan trips intelligently. The system provides destination recommendations, visa checking, budget estimation, safety advice, hotel suggestions, and itinerary planning using a knowledge base and inference rules. This project demonstrates the practical implementation of Artificial Intelligence and expert systems in solving real-world travel management problems efficiently.